



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2011 HSC Course Assessment Task 2

General instructions

- Working time – 60 minutes.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M3A – Mr Lam
- ☐ 12M3B – Mr Weiss
- ☐ 12M3C – Mr Law
- ☐ 12M4A – Mr Fletcher/Mrs Collins
- ☐ 12M4B – Mr Ireland
- ☐ 12M4C – Mrs Collins/Mr Rezcallah

STUDENT NUMBER # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{13}$	$\overline{12}$	$\overline{10}$	$\overline{9}$	$\overline{9}$	$\overline{53}$	

Question 1 (13 Marks)

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Marks(a) Differentiate each of the following with respect to x :

i. $x^3 e^{-2x}$ **2**

ii. $\frac{\ln x}{\cos x}$ **2**

iii. $\sin^2 3x$ **2**

iv. $\log_{10}(1-x)$ **2**

(b) Find:

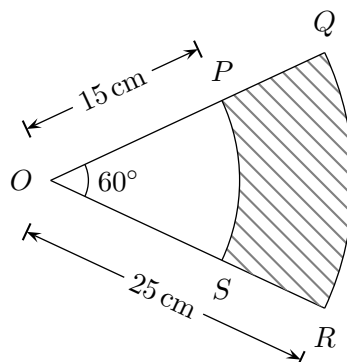
i. $\int \frac{e^{2x}}{e^{2x} + 1} dx$ **2**

ii. $\int \frac{dx}{(3x-1)^2}$ **1**

iii. $\int \tan x dx$ **2**

Question 2 (12 Marks)

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Marks(a) State the period and amplitude of $y = \frac{1}{3} \sin \left(2x - \frac{\pi}{4} \right)$. **2**(b) Solve $8 \sin \theta \cos \theta \tan \theta = \operatorname{cosec} \theta$ for $0 \leq \theta \leq 2\pi$. **3**(c) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin 2x}{\tan 3x} \right)$. **2**(d) PS and QR are arcs of concentric circles with centre O .
Calculate in terms of π ,i. the area of the shaded region $PQRS$. **2**ii. The perimeter of the shaded region $PQRS$. **3**

Question 3 (10 Marks)

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Marks

- (a) Solve the equation $\ln(x+6) + \ln(x-3) = \ln 5 + \ln 2$ **3**
- (b) i. Differentiate $x \cos x$. **1**
- ii. Hence or otherwise, evaluate $\int_0^{\frac{\pi}{3}} x \sin x \, dx$. **3**
- (c) Find the equation of the tangent to the curve $y = e^{\tan x}$ at the point on the curve where $x = \frac{\pi}{4}$. **3**

Question 4 (9 Marks)

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Marks

- (a) Solve the equation $e^{6x} - 7e^{3x} + 6 = 0$. **3**
- (b) Find $\frac{dy}{dx}$ if $y = \ln\left(\frac{x^4}{1-x^3}\right)$. **2**
- (c) Use mathematical induction to show for $n \geq 1$, **4**

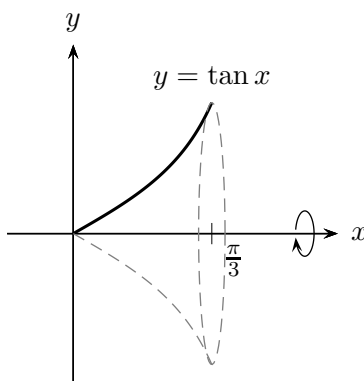
$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \cdots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$$

Question 5 (9 Marks)

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Marks

- (a) The area between the curve $y = \tan x$, the x axis and the ordinates $x = 0$ and $x = \frac{\pi}{3}$ is rotated about the x axis. **3**



Find the exact volume of the solid which is generated.

- (b) i. On the same set of axes, sketch the graphs of $y = \sin x$ and $y = 1 - \cos x$ for $0 \leq x \leq \pi$. **2**
- ii. Find the values of x where $\sin x = 1 - \cos x$ for $0 \leq x \leq \pi$. **2**
- iii. Calculate the area enclosed the curves $y = \sin x$ and $y = 1 - \cos x$ for $0 \leq x \leq \pi$. **2**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested Solutions

Question 1

(a) i. (2 marks)

- ✓ [1] for correct application of product rule.
- ✓ [1] for final answer.

$$\begin{aligned} & \frac{d}{dx} (x^3 e^{-2x}) \\ u &= x^3 \quad v = e^{-2x} \\ u' &= 3x^2 \quad v' = -2e^{-2x} \\ & \frac{d}{dx} (x^3 e^{-2x}) \\ &= x^3 \cdot (-2e^{-2x}) + e^{-2x} \cdot 3x^2 \\ &= 3x^2 e^{-2x} - 2x^3 e^{-2x} \\ &= x^2 e^{-2x} (3 - 2x) \end{aligned}$$

ii. (2 marks)

- ✓ [1] for correct application of quotient rule.
- ✓ [1] for final answer.

$$\begin{aligned} & \frac{d}{dx} \left(\frac{\ln x}{\cos x} \right) \\ u &= \ln x \quad v = \cos x \\ u' &= \frac{1}{x} \quad v' = -\sin x \\ & \frac{d}{dx} \left(\frac{\ln x}{\cos x} \right) = \frac{\cos x \cdot \frac{1}{x} - \ln x \cdot \sin x}{\cos^2 x} \\ &= \frac{\frac{\cos x}{x} + \sin x \ln x}{\cos^2 x} \times \frac{x}{x} \\ &= \frac{\cos x + x \sin x \ln x}{x \cos^2 x} \end{aligned}$$

iii. (2 marks)

- ✓ [1] for correct application of chain rule.
- ✓ [1] for final answer.

$$\begin{aligned} & \frac{d}{dx} (\sin^2 3x) = 2 \sin 3x \times 3 \cos 3x \\ &= 6 \sin 3x \cos 3x \end{aligned}$$

iv. (2 marks)

- ✓ [1] for correctly changing base from 10 to e .
- ✓ [1] for correct differentiation.

$$\begin{aligned} \frac{d}{dx} (\log_{10}(1-x)) &= \frac{d}{dx} \left(\frac{\ln(1-x)}{\ln 10} \right) \\ &= \frac{1}{\ln 10} \times -\frac{1}{1-x} \\ &= \frac{1}{\ln 10(x-1)} \end{aligned}$$

(b) i. (2 marks)

- ✓ [-1] for each mistake.

$$\int \frac{e^{2x}}{e^{2x} + 1} dx = \frac{1}{2} \ln(e^{2x} + 1) + C$$

ii. (1 mark)

$$\begin{aligned} \int \frac{dx}{(3x-1)^2} &= \int (3x-1)^{-2} dx \\ &= -\frac{1}{3} (3x-1)^{-1} + C \\ &= -\frac{1}{3(3x-1)} + C \end{aligned}$$

iii. (2 marks)

- ✓ [-1] for each mistake.

$$\begin{aligned} \int \tan x dx &= \int \frac{\sin x}{\cos x} dx \\ &= -\int \frac{-\sin x}{\cos x} dx \\ &= -\ln(\cos x) + C \end{aligned}$$

Question 2

(a) (2 marks)

- ✓ [1] each for a and T .

$$\begin{aligned} y &= \frac{1}{3} \sin \left(2x - \frac{\pi}{4} \right) \equiv a \sin(nx + \phi) \\ a &= \frac{1}{3} \quad T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi \end{aligned}$$

(b) (3 marks)

✓ [1] for resolving $\tan \theta$ and $\operatorname{cosec} \theta$ into $\sin \theta$ and $\cos \theta$.✓ [1] for each of $\frac{\pi}{6}$, $\frac{5\pi}{6}$.

$$\begin{aligned}
 8 \sin \theta \cos \theta \tan \theta &= \operatorname{cosec} \theta \\
 8 \sin \theta \times \cancel{\cos \theta} \times \frac{\sin \theta}{\cancel{\cos \theta}} &= \frac{1}{\sin \theta} \\
 8 \sin^3 \theta &= 1 \\
 \sin^3 \theta &= \frac{1}{8} \\
 \therefore \sin \theta &= \frac{1}{2}
 \end{aligned}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

✓ [1] for final answer.

✓ [0] for $\ln \left(\frac{x+6}{x-3} \right) = \ln 10$ as this sidesteps the quadratic and also the need to discard $x = -7$.

$$\ln(x+6) + \ln(x-3) = \ln 5 + \ln 2$$

$$\ln[(x+6)(x-3)] = \ln(5 \times 2)$$

$$(x+6)(x-3) = 10$$

$$x^3 + 3x - 28 = 0$$

$$(x+7)(x-4) = 0$$

$$x = -7, 4$$

However, $\ln(-7+6)$ and $\ln(-7-3)$ are not defined.

$$\therefore x = 4$$

(c) (2 marks)

✓ [1] for manipulating original limit to become $\frac{2}{3} \times \frac{\sin 2x}{2x} \times \frac{3x}{\tan 3x}$. If this is not done, mark is not awarded.

✓ [1] for final answer.

$$\begin{aligned}
 \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{\tan 3x} \right) &= \frac{2}{3} \times \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \times \frac{3x}{\tan 3x} \right) \\
 &= \frac{2}{3} \times 1 \times 1 = \frac{2}{3}
 \end{aligned}$$

i. (1 mark)

$$\frac{d}{dx}(x \cos x)$$

$$u = x \quad v = \cos x$$

$$u' = 1 \quad v' = -\sin x$$

$$\therefore \frac{d}{dx}(x \cos x) = -x \sin x + \cos x$$

ii. (3 marks)

✓ [1] for $\int_0^{\frac{\pi}{3}} x \sin x dx = \int_0^{\frac{\pi}{3}} \cos x dx - \int_0^{\frac{\pi}{3}} \frac{d}{dx}(x \cos x) dx$

✓ [1] for finding the primitive of the two functions.

✓ [1] for final answer.

Since $\frac{d}{dx}(x \cos x) = -x \sin x + \cos x$,

$$\therefore x \sin x = \cos x - \frac{d}{dx}(x \cos x)$$

$$\therefore \int_0^{\frac{\pi}{3}} x \sin x dx$$

$$= \int_0^{\frac{\pi}{3}} \cos x dx - \int_0^{\frac{\pi}{3}} \frac{d}{dx}(x \cos x) dx$$

$$= \left[\sin x \right]_0^{\frac{\pi}{3}} - \left[x \cos x \right]_0^{\frac{\pi}{3}}$$

$$= \left(\frac{\sqrt{3}}{2} - 0 \right) - \left(\frac{\pi}{3} \cdot \frac{1}{2} - 0 \right)$$

$$= \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

(d) i. (2 marks)

$$\begin{aligned}
 A &= \frac{1}{2} \times 25^2 \times \frac{\pi}{3} - \frac{1}{2} \times 15^2 \times \frac{\pi}{3} \\
 &= \frac{200\pi}{3} \text{ cm}^2
 \end{aligned}$$

ii. (3 marks)

$$\begin{aligned}
 P &= PS + QR + PQ + RS \\
 &= \left(15 \times \frac{\pi}{2} \right) + \left(25 \times \frac{\pi}{3} \right) + 10 + 10 \\
 &= 20 + \frac{40\pi}{3} \text{ cm}
 \end{aligned}$$

Question 3

(a) (3 marks)

✓ [1] for $\ln[(x+6)(x+3)] = \ln(5 \times 2)$.✓ [1] for $x = -7, 4$.

(c) (3 marks)

$$y = e^{\tan x} \Rightarrow \frac{dy}{dx} = \sec^2 x e^{\tan x}$$

$$\text{At } x = \frac{\pi}{4},$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\cos^2 \frac{\pi}{4}} e^{\tan \frac{\pi}{4}} \\ &= (\sqrt{2})^2 \times e^1 \\ &= 2e \\ y &= e^{\tan \frac{\pi}{4}} = e \end{aligned}$$

By the point gradient formula,

$$y - e = 2e \left(x - \frac{\pi}{4} \right)$$

$$y = 2ex - \frac{e\pi}{2} + e$$

Question 4

(a) (3 marks)

- ✓ [1] for correct factorisation.
- ✓ [1] for $e^{3x} = 1$ and $e^{3x} = 6$.
- ✓ [1] for final solutions.

$$e^{6x} - 7e^{3x} + 6 = 0$$

$$\text{Let } m = e^{3x},$$

$$\begin{aligned} m^2 - 7m + 6 &= 0 \\ (m - 6)(m - 1) &= 0 \\ m &= 1, 6 \\ \therefore e^{3x} &= 1, 6 \\ e^{3x} = 1 &\quad \left| \quad e^{3x} = 6 \right. \\ 3x = 0 &\quad \left| \quad 3x = \ln 6 \right. \\ x = 0 &\quad \left| \quad x = \frac{1}{3} \ln 6 \right. \\ \therefore x &= 0 \text{ or } \frac{1}{3} \ln 6 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for using log rules to resolve fraction into a simpler expression.
- ✓ [1] for final answer.

$$y = \ln \left(\frac{x^4}{1 - x^3} \right) = \ln x^4 - \ln(1 - x^3)$$

$$\frac{dy}{dx} = \frac{4}{x} - \frac{-x^2}{1 - x^3} = \frac{4}{x} + \frac{x^2}{1 - x^3}$$

(c) (4 marks)

- ✓ [1] for correct evaluation of base case.
- ✓ [1] for correct assumption in inductive step.
- ✓ [2] for correct inductive step part 2.

$$\begin{aligned} \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots \\ + \frac{1}{(4n - 3)(4n + 1)} = \frac{n}{4n + 1} \end{aligned}$$

- Base case: $n = 1$

$$\begin{aligned} \frac{1}{1 \times 5} &= \frac{1}{5} \\ \frac{1}{4(1) + 1} &= \frac{1}{5} \end{aligned}$$

Hence the statement is true for $n = 1$.

- Inductive step: assume the statement is true for some value $k \in \mathbb{R}$, i.e.

$$\begin{aligned} \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots \\ + \frac{1}{(4k - 3)(4k + 1)} = \frac{k}{4k + 1} \end{aligned}$$

and evaluate the statement when $n = k + 1$:

$$\begin{aligned} &\overbrace{\frac{1}{1 \times 5} + \dots + \frac{1}{(4k - 3)(4k + 1)}}^{= \frac{k}{4k + 1}} \\ &+ \frac{1}{(4(k + 1) - 3)(4(k + 1) + 1)} \\ &= \frac{k}{4k + 1} + \frac{1}{(4k + 1)(4k + 5)} \\ &= \frac{k(4k + 5) + 1}{(4k + 1)(4k + 5)} \\ &= \frac{4k^2 + 5k + 1}{(4k + 1)(4k + 5)} \\ &= \frac{(4k + 1)(k + 1)}{(4k + 1)(4k + 5)} \\ &= \frac{k + 1}{4(k + 1) + 1} \end{aligned}$$

The statement is also true for $n = k + 1$. By the principle of mathematical induction, the statement is true for all positive integers n .

ii. (2 marks)

$$x = 0, \frac{\pi}{2}$$

iii. (2 marks)

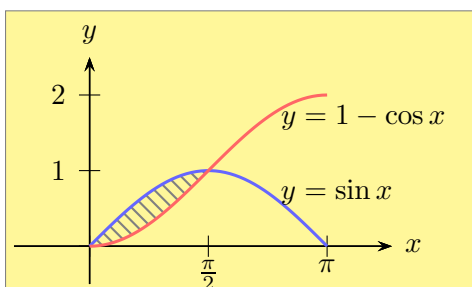
Question 5

(a) (3 marks)

$$\begin{aligned} V &= \pi \int y^2 dx = \pi \int_0^{\frac{\pi}{3}} \tan^2 x dx \\ &= \pi \int_0^{\frac{\pi}{3}} \sec^2 x - 1 dx \\ &= \pi \left[\tan x - x \right]_0^{\frac{\pi}{3}} = \pi \left(\tan \frac{\pi}{3} - \frac{\pi}{3} \right) \\ &= \pi \left(\sqrt{3} - \frac{\pi}{3} \right) = \pi \left(\frac{3\sqrt{3} - \pi}{3} \right) \end{aligned}$$

(b) i. (2 marks)

- ✓ [1] for $y = 1 - \cos x$.
- ✓ [1] for $y = \sin x$.
- ✓ [-1] if *any* of the following are true:
 - no labels (one label = no penalty)
 - no scale
 - larger domain than required
 - extra solution(s) to x



$$\begin{aligned} A &= \int_0^{\frac{\pi}{2}} \sin x - (1 - \cos x) dx \\ &= \int_0^{\frac{\pi}{2}} \sin x + \cos x - 1 dx \\ &= \left[-\cos x + \sin x - x \right]_0^{\frac{\pi}{2}} \\ &= \left(\sin \frac{\pi}{2} - \cancel{\sin 0} \right) \\ &\quad - \left(\cancel{\cos \frac{\pi}{2}} - \cos 0 \right) - \left(\frac{\pi}{2} - 0 \right) \\ &= 2 - \frac{\pi}{2} = \frac{4 - \pi}{2} \end{aligned}$$